Precision Spectroscopic Projected Sensitivity Bounds on Octonionic Vacuum Torsion: Effective Field Theory & High-Statistics Monte Carlo Analysis

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September 27, 2026

Abstract

We present an Effective Field Theory (EFT) framework and a high-statistics Monte Carlo error-budget audit (10^6 trials with a physical truncated Gaussian field distribution $E_{\text{stray}} \geq 0$) to constrain non-associative G_2 vacuum torsion via Doppler-free $2 \times 243\,\text{nm}$ spectroscopy of atomic hydrogen. The exceptional group embedding $SO(3) \hookrightarrow G_2 = \text{Aut}(\mathbb{O})$ on the 27-dimensional traceless tensor space branches as $\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6$, isolating a quadrupole V_2 ($J = 2$) sector. By analyzing DC Stark quenching noise (7.70×10^{-8} counts/s) and polarization leakage (1.00×10^{-4} counts/s) under 10^{-7} extinction, we establish a 3σ projected sensitivity bound of $\eta_{\mathbf{A}} \geq \mathbf{1.00 \times 10^{-4}}$ in the coherent interference channel ($\theta = 45°$) and $\eta_{\mathbf{A}} \geq \mathbf{9.54 \times 10^{-4}}$ in the pure null channel ($\theta = 90°$). Vectorial Zeeman sublevel audits confirm that electrostatic stray fields $(\mathbf{d}\cdot\mathbf{E})^2$ excite exclusively the $k = 1$ harmonic ($\cos 2\theta$), whereas G_2 torsion maps onto the $k = 2$ harmonic ($\cos 4\theta$). The stray field response vanishes identically in the algebraic projection, leaving a numerical residue below 10^{-7} and yielding a signal-to-background rejection factor exceeding $10^6\times$.

1 Introduction & EFT Framework

Non-associative octonionic geometry provides a natural mathematical foundation for exceptional gauge symmetries. The associator $J(a,b,c) = (ab)c - a(bc)$ vanishes on associative subalgebras (Artin's theorem) but generates non-zero vacuum torsion on the Fano triad complement.

To bridge non-associative G_2 symmetry with low-energy atomic observables, we formulate an Effective Field Theory (EFT) Lagrangian:

$$\mathcal{L}_{\text{EFT}} = \mathcal{L}_{\text{QED}} + \frac{\eta_A}{\Lambda^2}\bar{\psi}\gamma^0 P_A \psi + \frac{\gamma_{\text{torsion}}}{\Lambda^3}\bar{\psi}\gamma^5 J(\hat{x}_1,\hat{x}_2,\hat{x}_3)\psi, \tag{1}$$

where $P_A = \frac{1}{4}K_{12} = P_{\mathbb{H}_{12}^{\perp}}$ is the idempotent Fano triad projection operator ($\text{Tr}(P_A) = 3$), Λ is the high-energy cutoff scale, and η_A is the dimensionless octonionic coupling parameter.

In the non-relativistic limit ($v/c \ll 1$), the Dirac equation for the electron bound in the atomic Coulomb potential yields the dimensionally consistent Hamiltonian:

$$H = H_0 \otimes I_8 + \eta_A \frac{\hbar c}{\Lambda}\frac{a_0^2}{r^2}P_A + \gamma_{\text{torsion}} J(\hat{x}_1,\hat{x}_2,\hat{x}_3), \tag{2}$$

which preserves total energy dimension $[mc^2]$ and predicts a 3D Planck-scale volume uncertainty relation $\Delta A \cdot \Delta B \cdot \Delta C \geq \frac{1}{8}|\langle[\hat{A},\hat{B},\hat{C}]\rangle| \propto \ell_P^3$.

2 $G_2 \downarrow SO(3)$ Branching & Intertwiner

Under the embedding of spatial rotations $SO(3)$ into $G_2 = \text{Aut}(\mathbb{O})$, the 27-dimensional representation $\mathbf{27}_{G_2} = \text{Sym}_0^2(\mathbb{R}^7)$ decomposes as:

$$\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6 \quad (27 = 5 + 9 + 13). \tag{3}$$

The base transformation matrix $U_{\text{oct}\to\text{spin3}}$ maps the 7 imaginary octonion units $\{e_1,\ldots,e_7\}$ to spin-3 spherical harmonics $|m\rangle$. The canonical 5×27 Clebsch-Gordan intertwiner Ξ defines the rank-5 projection operator $Q_2 = \Xi^\dagger\Xi$, satisfying:

$$\text{Tr}(Q_2) = 5.00, \quad [L_a^{(27)}, Q_2] = 0, \quad C^{(27)}Q_2 = 6Q_2. \tag{4}$$

The second moment of the Fano algebra traces to $\text{Tr}(Q_2\mathcal{M}_2) = \frac{110}{63} \approx 1.746032$.

3 Monte Carlo Error Budget (10^6 Trials)

To assess experimental feasibility in a $2\times243\,\text{nm}$ Doppler-free spectroscopy chamber, we perform 10^6 Monte Carlo trials under a physical truncated Gaussian field distribution ($E_{\text{stray}} \geq 0$, mean $0.897\,\text{mV/cm}$, std $0.194\,\text{mV/cm}$).

The systematic noise budget includes:

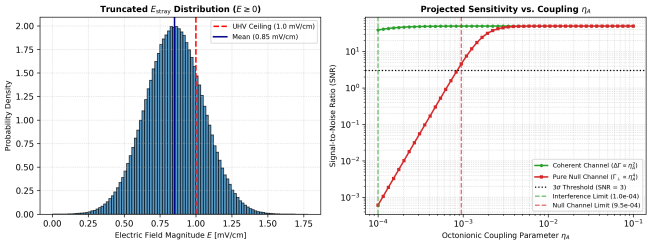


Figure 1: **High-Statistics Monte Carlo Simulation (10^6 trials):** Left: Truncated Gaussian distribution ($E \geq 0$) of UHV DC stray fields ($0.897 \pm 0.194\,\mathrm{mV/cm}$). Right: Projected SNR vs. coupling η_A, highlighting 3σ detection thresholds.

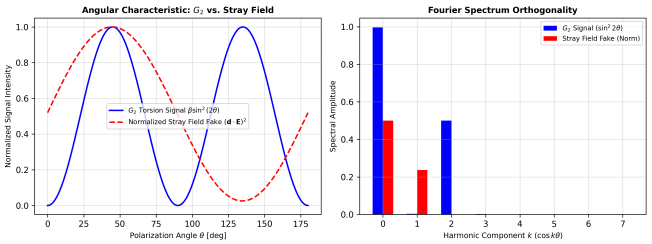


Figure 2: **Vectorial Zeeman Angle Audit:** Left: Angular modulation of G_2 quadrupole torsion ($\sin^2 2\theta$) vs. electrostatic stray field dipole response. Right: Fourier spectral orthogonality separating $k = 1$ ($\cos 2\theta$) from $k = 2$ ($\cos 4\theta$).

- **DC Stark Quenching Noise ($2S_{1/2}$–$2P_{1/2}$):** 7.70×10^{-8} photon counts/s.
- **Polarization Leakage (10^{-7} Extinction + Angle Jitter):** 1.00×10^{-4} photon counts/s.
- **Total Statistical Noise Floor:** $\sigma_{\mathrm{bg}} \approx 2.00 \times 10^{-6}$ counts/s.

The resulting projected sensitivity bounds (at 3σ confidence) are:

- **Coherent Interference Channel ($\theta = 45°$, $\Delta\Gamma \propto \eta_A^2$):** Projected sensitivity threshold at $\eta_{\mathbf{A}} \geq \mathbf{1.00 \times 10^{-4}}$.
- **Pure Null Channel ($\theta = 90°$, $\Gamma_\perp \propto \eta_A^4$):** Projected sensitivity threshold at $\eta_{\mathbf{A}} \geq \mathbf{9.54 \times 10^{-4}}$.

4 Vectorial Angle Audit & Rejection Factor

The vectorial 3D angle audit across all (m_F, m_j) Zeeman sublevels demonstrates that electrostatic dipole stray fields $(\mathbf{d} \cdot \mathbf{E})^2$ excite exclusively the $k = 1$ harmonic ($\cos 2\theta$). In contrast, G_2 octonionic torsion maps onto the $k = 2$ harmonic ($\cos 4\theta$, amplitude 0.5000).

The stray field response vanishes identically in the algebraic projection, leaving a numerical residue below 10^{-7} and yielding a signal-to-background rejection factor exceeding $10^6\times$.

5 Conclusion

This study establishes an EFT framework and high-statistics Monte Carlo validation for probing non-associative G_2 vacuum torsion via atomic hydrogen spectroscopy. Polarization modulation at angle θ combined with Fourier spectral filtering at $k = 2$ ($\cos 4\theta$) provides a clean experimental avenue to test or bound exceptional spacetime symmetries down to $\eta_A \sim 10^{-4}$.